CLASS 6 — MATHS · FORMULA SHEET

Chapter 9: **Symmetry** | Quick revision, one page

Line (reflection) symmetry

- A line of symmetry divides a figure into two parts that fit exactly on each other when folded (mirror halves).
- Mirror image: a point is as far behind the mirror line as the original is in front of it.
- A figure may have 0, 1, 2, 3, ... or infinitely many lines of symmetry (circle).

Rotational symmetry

- A figure has rotational symmetry if it fits onto itself after a turn of less than 360° about a point (the centre).
- Order of rotational symmetry $=$ number of times it fits onto itself in one full turn.
- Smallest angle of rotation $= \dfrac{360^\circ}{\text{order}}$.

Regular polygons

- A regular polygon with n sides has n lines of symmetry and rotational symmetry of order n.
- Equilateral triangle: 3 lines, order 3, angle 120°. Square: 4 lines, order 4, angle 90°.
- Regular hexagon: 6 lines, order 6, angle 60°. Circle: infinite lines and infinite order.

Other shapes and letters

- Rectangle (not square): 2 lines, order 2. Rhombus (not square): 2 lines, order 2. Parallelogram: 0 lines, order 2.
- Isosceles triangle: 1 line, order 1. Scalene triangle: 0 lines, order 1.
- Letters: H, O, X: 2 lines and order 2. A, T, M, U: 1 line. S, N, Z: no line but order 2.

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